

BHANU TEJA GULLAPALLI

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PROFESSIONAL SUMMARY

Machine learning scientist with 9+ years of experience developing and deploying personalized AI systems for digital health, wearable sensing, and sequential decision-making. Skilled in reinforcement learning, online learning, Bayesian modeling, time-series analysis, and deep learning, with hands-on experience in production AI deployment, model evaluation, monitoring, retraining pipelines, and LLM-enabled development for real-world clinical studies.

TECHNICAL SKILLS

AI & Machine Learning: Reinforcement Learning, Online Learning, Bayesian Modeling, Statistical Modeling, Statistical Analysis, Survival Analysis, Sequential Decision Making, Time-Series Modeling, Deep Learning, Uncertainty Quantification, Transformers, Gaussian Processes

Production AI Systems: Production ML Systems, REST APIs, GCP, Docker, PostgreSQL, Monitoring & Alerting, Logging, Load Testing, Model Deployment, Model Retraining Workflows, Reproducibility Frameworks

AI Development Tools: LLMs, Agentic Coding Workflows, LLM-Assisted Development

Frameworks & Tools: Python, PyTorch, Scikit-learn, TensorFlow, Hugging Face, Pandas, NumPy, PySpark, Linux, Git, Bash

EXPERIENCE

Postdoctoral Researcher, Harvard, Boston, MA

July 24 – Current

- Developed algorithms to personalize when digital health interventions were delivered, improving timing-evaluation performance by 15% and better aligning interventions with participant behavior patterns.
- Deployed resulting AI decision-making platform to support approximately 55,000 sequential intervention decisions across 200 participants during a 135-day randomized controlled trial.
- Built backend APIs, databases, Dockerized services, and model retraining workflows supporting daily AI-driven intervention delivery with approximately 96% operational reliability.
- Created monitoring, alerting, logging, fallback, validation, and recovery systems, reducing production risk while enabling recovery of affected data.
- Established reproducibility frameworks for online learning systems; reproduced intervention-timing performance across offline testing and live deployment, co-authored a peer-reviewed workflow paper, and led adoption of agentic AI coding workflows.

Research Intern, Optum AI Labs

June 23 – September 23

- Built CGM-based risk-estimation models for Type 2 Diabetes Mellitus, validating early glucose-spike prediction on public data from 109 T2DM participants and a private dataset of 3,870 participants.
- Created multimodal time-series models using continuous glucose, glucose rate-of-change, and temporal features, achieving 0.71 F1-score, 0.70 precision, and 0.76 recall for early spike-onset prediction.

¹Use URL bhanutejagullapalli.github.io in case hyperlinks don't work

- Designed spike-forecasting and sampling-based intervention strategies to support just-in-time walking interventions, achieving 37.6% MAPE and Spearman correlation of 0.41 for spike-count forecasting.

Research Intern, Samsung Research America, Mountain View, CA May 22 – September 22

- Designed a transfer-learning framework using temporal convolutional networks to extract heart-rate variability features from noisy earbud PPG signals collected from 18 human subjects.
- Improved HRV feature estimation by approximately 26% over HeartPy using Biopac reference signals as ground truth.
- Improved downstream stress-detection accuracy, sensitivity, and specificity by at least 5% using model-derived HRV features from consumer-grade earbud sensors.

Software Engineer, Samsung R&D Institute, Bangalore, India July 15 – December 16

- Shipped multimedia features for Samsung Galaxy video editing applications, including highlight generation, slow-motion editing, automated story generation, and theme-based video editing workflows.
- Collaborated with mobile systems, design, and UI teams to optimize performance and user experience for production Android devices, including Samsung Galaxy S8 Video Editor Pro.

EDUCATION

Ph.D. in Data Science, University of California, San Diego September 18 - June 24

Dissertation focused on wearable sensing, digital biomarkers, and deep learning models for detecting stages of the addiction cycle, including craving, drug use, and euphoria. Supported by NSF and Optum Labs grants.

Advised by *Prof. Tauhidur Rahman*

M.S. in Computer Science, University of Massachusetts, Amherst Feb 17 - Sept 18

B.Tech. in Computer Science, Indian Institute of Technology, Guwahati July 11 - May 15

SELECTED PUBLICATIONS (4 OF 17)

- Reproducible Workflow for Online AI in Digital Health
Bhanu Teja Gullapalli, Sushobhan Ghosh, Daiqi Gao, Asim Gazi, Anna Trella, and Susan A. Murphy
Philosophical Transactions of the Royal Society, 2026
- SigmaScheduling: Uncertainty-Informed Scheduling of Decision Points for Intelligent Mobile Health Interventions
Bhanu Teja Gullapalli, Asim Gazi, Daiqi Gao, Benjamin M. Marlin, Vivek Shetty, and Susan A. Murphy
IEEE International Conference on Body Sensor Networks (BSN), 2025
- Opioid Misuse Detection from Cognitive and Physiological Data with Temporal Fusion Deep Learning
Bhanu Teja Gullapalli, Yunfei Luo, Tauhidur Rahman, and Eric L. Garland
Drug and Alcohol Dependence, 2025
- Pharmacokinetics-Informed Neural Network for Predicting Opioid Administration Moments with Wearable Sensors
Bhanu Teja Gullapalli, Eric L. Garland, Stephanie Carreiro, Brittany Chapman, and Tauhidur Rahman
IAAI, AAAI Conference on Artificial Intelligence, 2024